

The Transition to College for Gender Minority Youth: Survey Evidence from California

Objective

The transition from high school to college can be a challenge for students learning to dually navigate both new academic and social environments. While the literature has often drawn attention to the emergent differences across some subgroups (e.g., by race/ethnicity, income, first generation status), limited research considers the significant diversity within and across the binary dichotomy of gender identity, inadvertently concealing their unique experiences, particularly of transgender and nonbinary students (Feldman et al. 2022; Glavinic 2010; Goldberg and Kuvalanka 2018). . Therefore, it is critical to consider the differences that may emerge across gender identity during the high school to college transition.

This work thus aims to extend the literature on the experiences of gender minority students during matriculation in several ways. Drawing on the results of a large-scale survey, we utilize regression and principal component analysis (PCA) to explore patterns in high school experiences and college aspirations among gender minority students as well as investigate the challenges they face as they consider the transition to college.

Theoretical Background and Prior Literature

Research documents the differences in educational outcomes for gender minority youth compared to their counterparts. Broadly, evidence indicates that LGBTQ+ students, particularly transgender youth, are less likely to graduate from high school as well as aspire to, apply to, and enroll in college (Feldman et al. 2022; Greytak et al. 2009; Wilkinson et al. 2021), differences that persist even after controlling for demographic characteristics, family background, and state- and school-level fixed effects (Sansone 2019). It is important to note, however, that the transition to college is shaped by a confluence of institutional, interpersonal, and intrapersonal factors that can affect students' educational aspirations and attainment (Kurlaender et al. 2019). Collectively, the interaction of these factors may influence how students traverse the college-going process. However, for gender minority youth in particular, these factors are also mediated by stressors that can underlie societal conditions; in this case, both external (e.g., discrimination and violence) and internal stressors (e.g., expectations of prejudice, concealment, and internalized stigmas). These stressors, heightened for gender minority students, can have an effect on psycho-social outcomes (Hendricks and Testa 2012; Hunter et al. 2021; Meyer 2003), ultimately influencing their educational aspirations and navigation through the transition from high school to college. Given these stressors and the aforementioned factors that guide students' educational trajectories, one might expect such experiences to negatively impact the plans of gender minority youth for postsecondary enrollment.

On the other hand, we might similarly expect the converse to be true given that resilience is also an essential element of the minority stress framework (Meyer 2015). In fact, gender minority students often develop coping skills through access to LGBTQ+ (lesbian, gay, bisexual, transgender, and queer/questioning) community spaces and other supportive resources (Feldman et al. 2022; Heck et al. 2014), which can lead to collective action, a sense of purpose, and personal growth (DiFulvio 2011; Riggle et al. 2011). Moreover, gender minority youth that struggled socially in high school may look towards college as a new opportunity to fit in, while those who did not struggle as much may have additional worries. Similarly, LGBTQ+ students that plan to live away from home while in college may feel increased trepidation (i.e., leaving familial support) or, in contrast, relief (i.e., escaping familial discomfort or hostility).

Data and Methods

Survey Instrument and Sample Characteristics

In May 2023, the authors partnered with the California Student Aid Commission (CSAC) to document the experiences of high school seniors who intended to enroll in college in the fall. The web-based survey was sent to all seniors statewide that had submitted either the Free Application for Federal Student Aid (FAFSA) or the California Dream Act Application (CADAA) for the 2023-24 academic year. Table 1 reports the demographic characteristics of our analytical sample.

To collect data on gender identity, we adopted the “two-step” approach recommended by the Williams Institute given both its high sensitivity and specificity as well as increased reliability in accurately capturing transgender respondents (Saperstein and Westbrook 2021; The GenIUSS Group 2014). This approach includes two questions: (1) *What is your sex assigned at birth, on your original birth certificate?*, with ‘male’ and ‘female’ as response options; and (2) *What is your current gender identity?*, with ‘woman’, ‘man’, ‘nonbinary’, ‘prefer not to say’, and ‘other’ as response options. Respondents who chose ‘other’ also had the option to write in a free response text box. Using these two survey items, we classify respondents as cisgender or binary transgender. To measure students’ sexual orientation, we adapted the survey instrument to better capture the breadth of identities of students with a survey question that asked students to self-identify their sexual orientation through the following options: lesbian or gay; straight; bisexual; asexual; prefer not to say; and other, which also included the opportunity to write in a free response box (The GenIUSS Group, 2014). Additionally, race/ethnicity and parental level of education were also self-reported using pre-defined survey options.

Data Analysis

We examine a series of closed-ended Likert scale questions to understand if gender minority students' worries about the transition to college are distinct from those of their peers through regression analysis coupled with principal component analysis (PCA). First, we identify the number of dimensions of students’ worries about the college transition using PCA, including all questions related to worries about college in the survey. Our analysis revealed three distinct

constructs to students' worries about college, which we later detail in the findings.¹ We then used regression analysis to estimate the differences in each construct of worry by gender identity. We estimate three different specifications: (1) an unconditional model with the gender categories alone; (2) a model controlling for student demographics; and (3) a model that controls for both student demographics and a construct of students' reported high school experiences. The high school experience variable is an index based on a series of questions students were asked in the survey about their high school academic, social, advising, and college preparation experiences. As we found strong reliability² for one construct of high school experience, we include it as a linear continuous variable in the regression model.³

Findings

Results reveal that while most students, regardless of gender identity, planned to enroll in a four-year college or university, binary transgender and nonbinary students planned to enroll at increased rates compared to their peers (see Figure 1). Similarly, findings indicate that most students planned to complete a four-year degree; however, differences emerge by gender identity in students' plans to earn advanced degrees (see Figure 2). Students were also asked a number of Likert-scale questions about their high school experiences, including academic experiences, social experiences, interactions with adults in schools, and preparation for college. We constructed an index for those experiences and compared average high school experiences by gender categories⁴ (see Table 2). We find a statistically significant difference between the high school experiences of cisgender students and gender minority students, who report less positive experiences overall (see Figure 3). Moreover, results from a PCA indicate concerns about college can be divided into three constructs: (1) general worries, (2) worries about discrimination, and (3) worries about financial burdens (summary statistics of the PCA constructs are reported in Table 3). Figures 4 through 6 present ordinary least squares (OLS) regression estimates for each component of worry identified by gender identity, both with and without controls for high school experiences and student demographics. Among gender minority students, we find no statistical differences in general worries between different gender identities (see Table 4). Conversely, differences emerge related to worries about discrimination; while gender minority students maintain higher levels of worry about discrimination overall,

¹ We select components with eigenvalues larger than 1.0, which identifies the first three components. We then construct a worry index for each selected component for every student as a weighted sum of the student's answers to the battery of questions about college worries with weights from the eigenvector of the component.

² We check the consistency of student answers to high school experience questions using Cronbach's alpha ($\alpha > 0.7$).

³ We tested for alternative functional forms for the high school experience variable.

⁴ The high school experience index was constructed by adding the values of each item. All items range from -2 to 2, with -2 indicating a very negative experience, 2 indicating a very positive experience, and 0 indicating a neutral experience. The construct of the overall high school experience ranges from -12 to 12 and applies to students who reported non-missing values for all six high school experience items.

transgender students are more likely to report concerns about discrimination compared to nonbinary and gender diverse students (see Figure 5).

Significance

This analysis is positioned to extend the literature in several ways. First, we provide direct evidence on the experiences of LGBTQ+ students prior to college matriculation, leveraging survey data collected from a large sample of college-bound high school seniors with a special focus on documenting the supports, challenges, and disparities facing gender minority youth. Additionally, our results contribute more broadly to the discussion of gender identity, as it considers the diversity within and across the binary dichotomy of gender that education research tends to privilege. Moreover, we find that while gender minority students have heightened concerns about discrimination prior to enrolling in college, they still plan to enroll at higher rates than cisgender students—a potential reflection of the resilience of this population in the pursuit of their goals (Meyer, 2015). These results emphasize the importance of identifying how experiences may differ across gender identities in order to develop and strengthen support systems for students navigating the transition from high school to college. While this work is limited in its ability to clarify the policies and practices that may have supported or hindered gender minority students, research must continue to examine the nuanced identities across gender in order to better understand and support the educational futures of students with different lived experiences.

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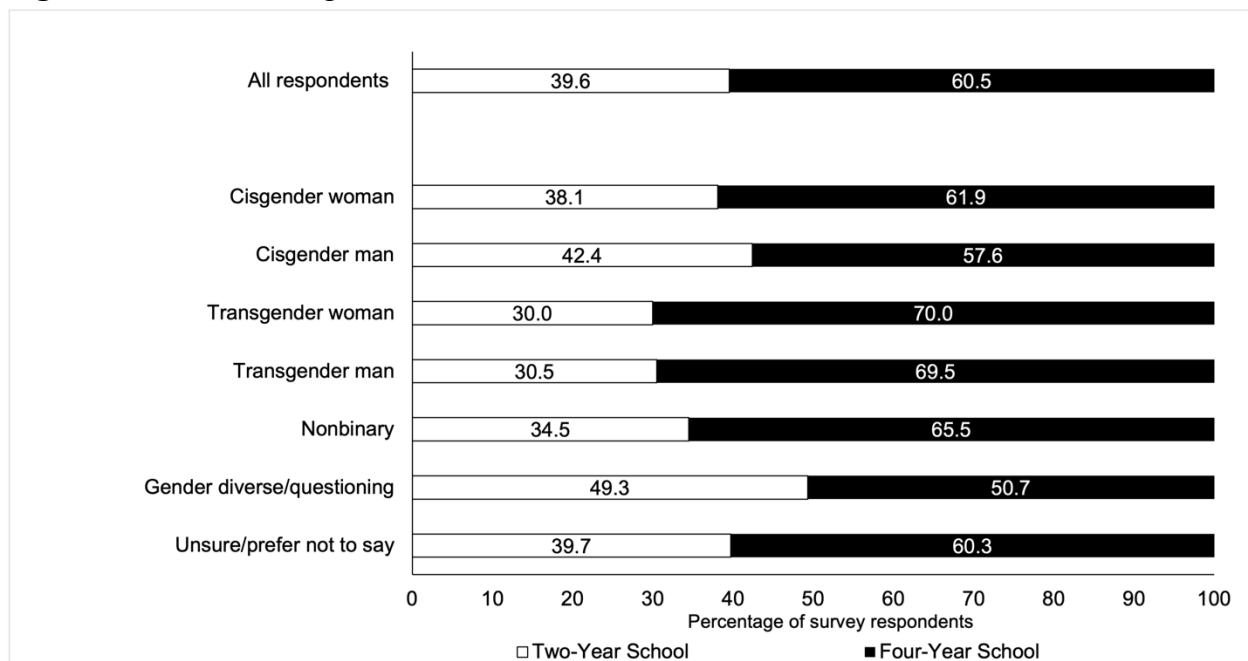
Tables and Figures

Table 1. Summary Statistics

	Freq.	Percent
Panel A. Gender identity		
Cisgender man	2,675	35.56
Cisgender woman	4,289	57.02
Transgender man	59	0.78
Transgender woman	20	0.27
Nonbinary	267	3.55
Gender diverse/questioning	71	0.94
Prefer not to say	141	1.87
Total reported gender identity	7,522	100.00
Panel B. Sexual orientation		
Straight/heterosexual	5,190	69.38
Gay or lesbian	330	4.41
Bisexual/pansexual/omnisexual	1,139	15.23
Asexual/aromantic/demisexual	182	2.43
Other/queer/questioning	131	1.75
Prefer not to say	509	6.80
Total reported sexual orientation	7,481	100.00
Panel C. Race/ethnicity		
American Indian/Alaskan Native	163	2.14
Black/African American	532	6.99
Hispanic/Latinx	3,793	49.82
Pacific Islander	64	0.84
Filipino	235	3.09
Asian	1,146	15.05
Other	182	2.39
White/Non-Hispanic	1,498	19.68
Total reported race/ethnicity	7,613	100.00
Panel D. Highest level of education among parents		
Did not complete high school	1,373	18.11
High school diploma	1,578	20.81
Some college, no college degree	975	12.86
Associate degree	452	5.96
Bachelor's degree	1,363	17.97
Graduate/Professional degree	1,455	19.19
Don't know	387	5.10
Total reported parental education	7,583	100.00

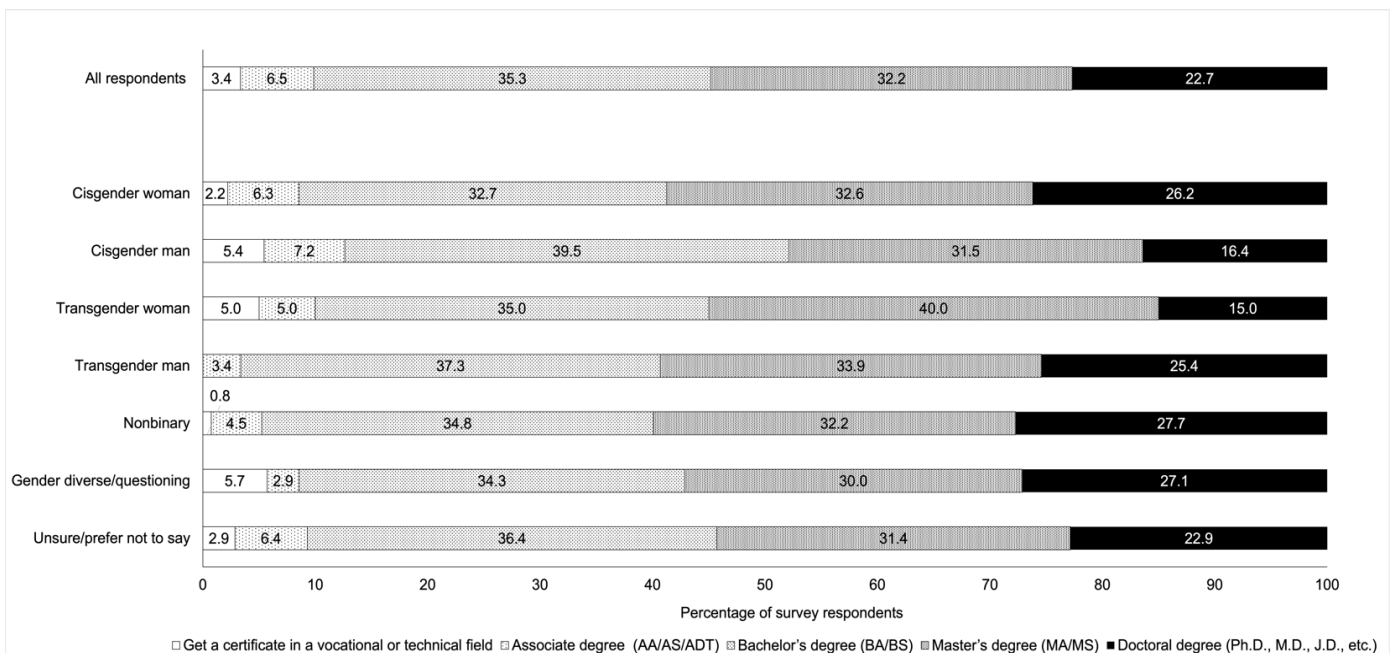
Note. Our sample only includes college-going high school seniors. Of the total sample ($n = 9,230$), 1,708 respondents did not report gender identity; 1,749 did not report sexual orientation; 1,617 did not report race/ethnicity; and 1,647 did not report level of parental education.

Figure 1. Plans for college enrollment



Note. Rates depicted in the figure represent 7,522 survey respondents who indicated plans to attend college. Of the full survey sample of high school seniors ($n = 10,221$), 9,230 respondents indicated that they planned to go to college, 170 said they did not plan to attend college, 294 expressed that they were unsure, and 527 did not respond to the question.

Figure 2. Plans for degree completion

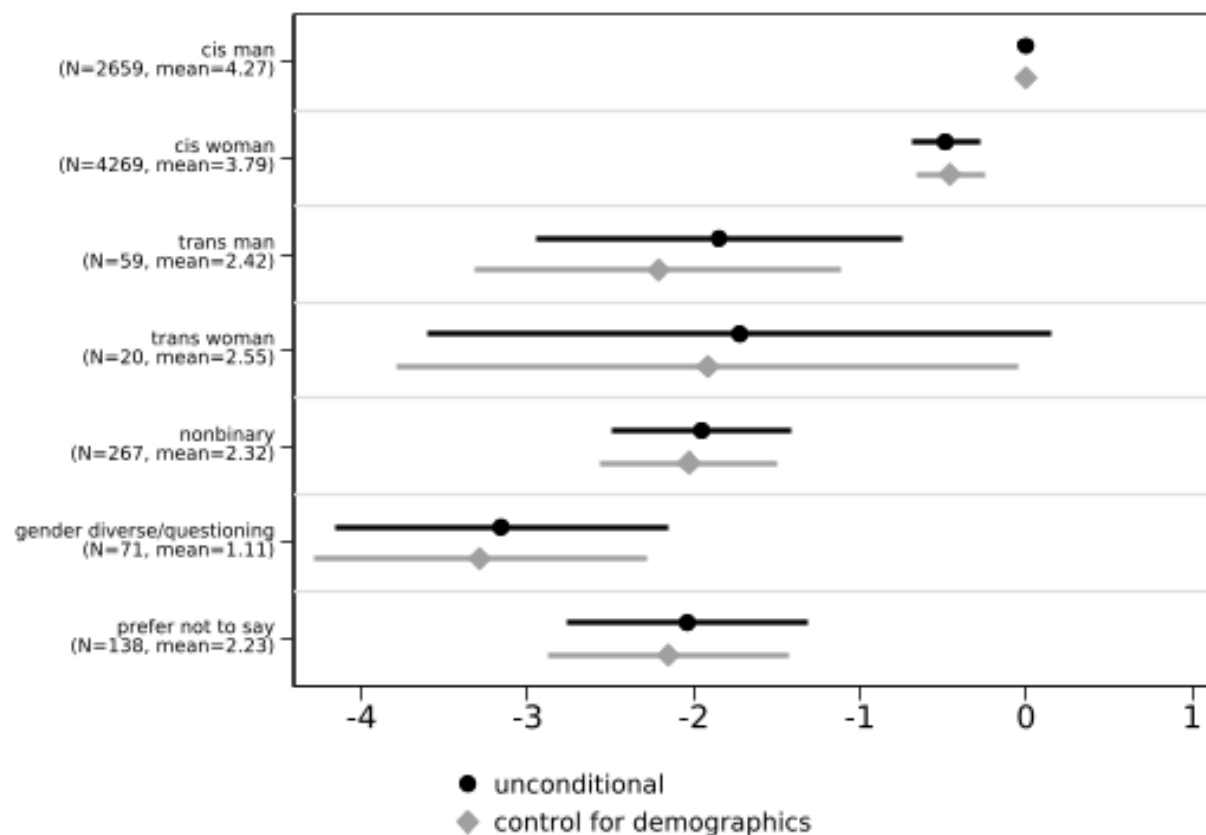


Note. Rates depicted in the figure represent 7,497 survey respondents who indicated plans for degree completion.

Table 2. General High School Experience by Gender Identity

	Freq.	General high school experience index
Cisgender man	2,659	4.271
Cisgender woman	4,269	3.787
Transgender man	59	2.424
Transgender woman	20	2.55
Nonbinary	267	2.318
Gender diverse/questioning	71	1.113
Prefer not to say	138	2.232
Total	7,483	3.839

Note. The general high school experience index only applies to students who reported non-missing values for all six high school experience items and ranges from -12 to 12 (negative values indicate a negative experience, positive values indicate a positive experience, and zero indicates a neutral experience).

Figure 3. Regression of Overall High School Experience on Gender Identity Categories

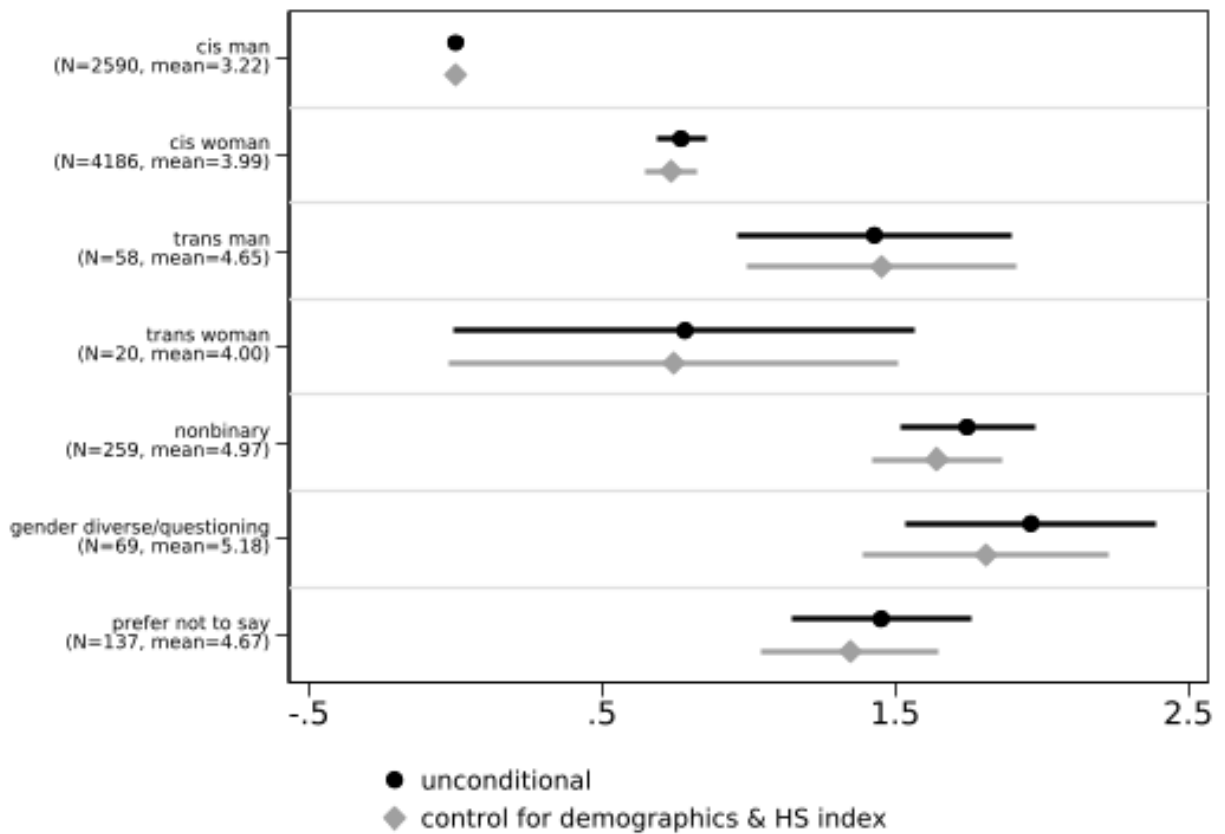
Note. For the unconditional model, $n = 7,483$. For the model controlling for demographics (race/ethnicity and parental level of education), $n = 7,464$. The lines indicate 95 percent confidence intervals while the sample frequency and mean index of each gender identity are reported in parentheses.

Table 3. College Concern Items and Results from Principal Component Analysis (PCA)

	N	Mean
Items:		
How worried are you about affording tuition fees	7,957	1.827
How worried are you about living expenses	7,956	1.676
How worried are you about performing well academically	7,958	1.502
How worried are you about balancing work obligations	7,943	1.557
How worried are you about balancing family obligations	7,941	1.055
How worried are you about finding community	7,934	1.316
How worried are you about living away from home	7,948	.908
How worried are you about having adequate support for emotional and mental health needs	7,944	1.127
How worried are you about discrimination based on gender	7,886	.436
How worried are you about discrimination based on sexual orientation	7,876	.333
How worried are you about discrimination based on race	7,895	.514
How worried are you about discrimination based on religion	7,891	.232
PCA Constructs:		
General worries (32.03%)	7,706	3.776
Worries about discrimination (13.79%)	7,706	-1.219
Worries about financial burdens (10.26%)	7,706	.634

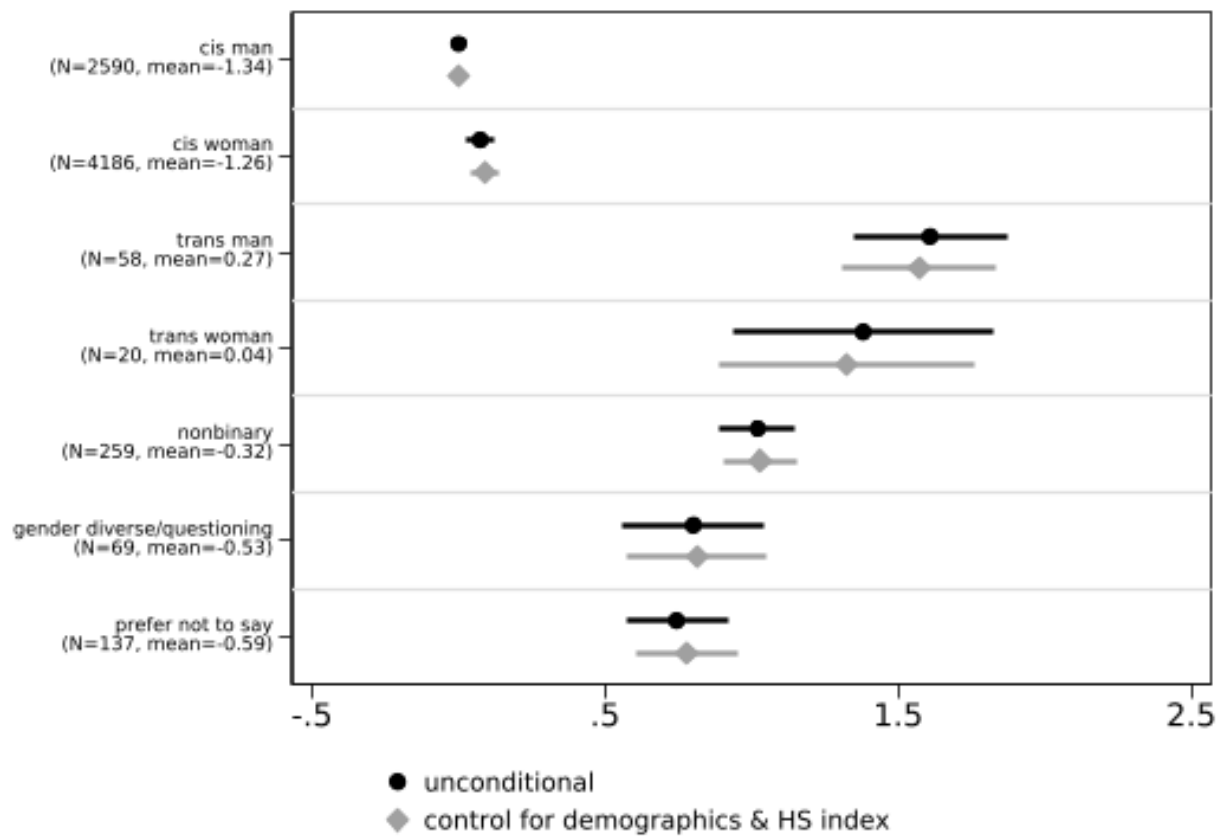
Note. The first twelve rows are items related to college worries included in the survey. Each item ranges from 0 to 3, where 0 represents not worried at all and 3 represents very worried. The last three rows indicate the three constructs: (1) *general worries* ranges from 0 to 10.23; (2) *worries about discrimination* ranges from -3.83 to 5.12; and (3) *worries about financial burdens* ranges from -3.64 to 3.65. Constructs are only created for high school students that reported non-missing values for all twelve items regarding college worries. The PCA proportion explained is reported in parentheses after each construct.

Figure 4. Regression of General College Worries on Gender Identity Categories



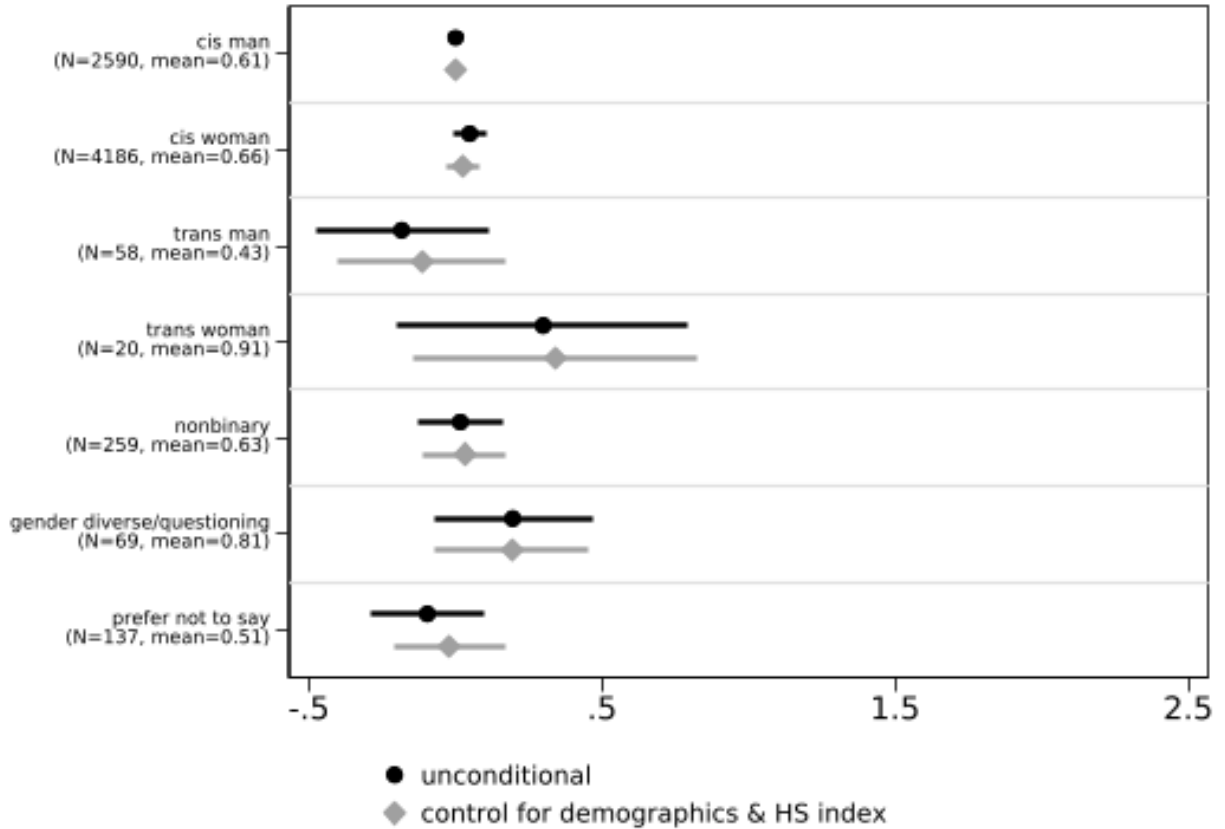
Note. We estimate regression models with gender indicators for an unconditional model ($n = 7,319$) and include race/ethnicity indicators, parental education indicators, and a constructed high school experience index as covariates for the specification with control variables ($n = 7,276$). The lines indicate 95 percent confidence intervals while the sample frequency and mean for each gender identity are reported in parentheses.

Figure 5. Regression of Worries about Discrimination on Gender Identity Categories



Note. We estimate regression models with gender indicators for an unconditional model ($n = 7,319$) and include race/ethnicity indicators, parental education indicators, and a constructed high school experience index as covariates for the specification with control variables ($n = 7,276$). The lines indicate 95 percent confidence intervals while the sample frequency and mean for each gender identity are reported in parentheses.

Figure 6. Regression of Worries about Financial Burdens on Gender Identity Categories



Note. We estimate regression models with gender indicators for an unconditional model ($n = 7,319$) and include race/ethnicity indicators, parental education indicators, and a constructed high school experience index as covariates for the specification with control variables ($n = 7,276$). The lines indicate 95 percent confidence intervals while the sample frequency and mean for each gender identity are reported in parentheses.